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MILITARY SPECIFICATION

IGNITION ELEMENT, ELECTRIC, MARK 10 MOD 0

(This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.)

1. SCOPE

1.1 This specification governs the manufacture, assembly and preparation for delivery of the Ignition Element, Electric, Mark 10 Mod 0, and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications, of the issue in effect on the date of invitation for bids, form a part of this document.

SPECIFICATIONS

FEDERAL

T-T-881	Twine, Cotton, Seine
QQ-T-425	Tinplate (Hot Dip and Electrolytic)
UU-C-282	Chipboard, Plain
LLL-F-291	Fiberboard, Corrugated, Single Face (Flexible)

MILITARY

MIL-C-10464	Can, Hermetic Sealing, Metal, Light Gage, Tear Strip Type
JAN-P-108	Packaging and Packing for Overseas Shipment Boxes, Fiberboard (U-board and W-board) Exterior and Interior

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1375

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-352	Forty Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	Transportation-Vibration Test for Use in Production of Fuzes
MIL-STD-354	Temperature and Humidity Test for Use in Production of Fuzes
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD No. 546641	Ignition Element, Electric, Mk 10 Mod 0
BUWEPS LD No. 546651	Transportation Vibration Test Fixture, Ignition Element
BUWEPS LD No. 546657	Firing Test Fixture, Ignition Element, (Output and Electrical Functioning)
BUWEPS LD No. 546658	Drop Test Fixture, Ignition Element
BUWEPS LD No. 546664	Packaging, Inner Container
BUWEPS LD No. 482723	Packing, Outer Container

And all documents listed thereon.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. -The following publication forms a part of this document. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission regulations are now a part of the Federal Regulations (1949 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78, Rev. 1950".)

3. REQUIREMENTS

3.1 General. -The ignition element shall be manufactured, assembled, and prepared for delivery each in accordance with its L. D. No. 546641 and all documents listed thereon.

3.1.1 Cleanliness and condition of parts. -All parts shall be free of chips, dirt, grease, rust and all other foreign materials. There shall be no burrs or other imperfections of manufacture which would harmfully affect either the assembly or the functioning of the product. The cleaning methods which are used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents which are used.

3.1.2 Materials. -The materials which are used in the fabrication of the ignition element components, and in the assembly of the ignition elements, shall be as specified in the applicable documents, except where deviation is specifically authorized by the Government.

3.1.3 Assembly procedure. -Assembly shall proceed in a careful, neat and orderly manner to prevent either damage to parts, or the inclusion of foreign materials and/or defective or marginal parts and subassemblies.

3.1.4 Preproduction samples. -Unless otherwise specified by the procuring activity, a preproduction sample of 500 Ignition Elements of one mark and mod shall be furnished the Government at the start of either pilot production, or full production if there is no pilot production. These ignition elements shall be manufactured by the same procedures and processes, and at the same location, which are proposed by the contractor for full production and using the same tooling, fixtures and equipment as proposed for full production. The preproduction sample shall be prepared for delivery as prescribed in the applicable section of this document. The preproduction sample shall conform to the requirements of this document as prescribed in 4.2.2. The Bureau of Naval Weapons shall designate the testing activity.

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3.2 Bridge wire assembly, continuity. -Immediately after soldering the bridge wire in place, the bridge wire assembly shall be thoroughly cleaned to remove solder flux and contamination. The bridge wire shall then be caused to glow by the application of an electric current to assure continuity of the bridge wire. The current shall be regulated to produce a visible red glow of both legs of the bridge wire. An EMF of approximately 1.5 volts AC or DC may be used with a suitable series resistor to adapt the generator characteristics to the resistance of the ignition element. Glow testing is a necessary manufacturing operation, and shall be conducted on each assembly. Assemblies which fail to glow in both legs of the bridge wire shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge wire.

3.3 Closing disc. -The closing disc shall be free from splits, cracks or any other deleterious imperfections of manufacture. It shall not be perforated, after the assembly operation.

3.4 Explosive Components, Volatile. -The primer composition (Frankford Arsenal FA-878, Dwg. B-7553516) used in the loading of the ignition element assembly shall have a moisture content of 3% at the time of loading. After consolidating the primer composition in the spacer, Dwg. 2302016 (loaded), it shall be dried at a temperature of 100° F to 120° F for 16 hours.

3.5 Igniter assembly, loading. -The consistency of the ignition charge and the loading procedure shall be regulated to secure an ignition charge which is free from internal voids, while avoiding damage to the bridge wire. The ignition charge shall fill the cavity such that the charge is flush with the top of the cavity and the bridge wire is completely covered. All lateral surfaces which are subject to metal contact shall be free from any of the ignition charge. Where necessary, lateral surfaces may be cleaned of ignition charge by means of a cloth wet with butyl acetate. The charged igniter assembly shall be dried at a temperature of 100° F to 120° F for a minimum of 12 hours. As least 4 hours of drying shall immediately precede assembly of the igniter assembly into the ignition element assembly. The dried igniter charge shall be free from cracks, voids, and separation from the metallic housing.

3.5.1 Ignition charge. -The ignition charge shall be compounded by an approved method which maintains the weight composition of non-volatile constituents prescribed by the drawing. The finished ignition charge, when ready for buttering into the igniter assembly, shall be a uniformly mixed slurry in which all the particles are wetted. The consistency shall be such that the liquid will not

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separate from the mixture when the dish is tilted. Adjustment of the consistency may be obtained by the addition of pure butyl acetate or by evaporation of the solvent. Any adjustment shall be followed by thorough remixing. An advisory procedure for preparing and loading the ignition charge is given in 6.3.

3.6 Insulation resistance. -When tested electrically, as prescribed in Para. 4.4.1, the insulation resistance of the Inner Cup and Electrode Assembly shall be in accordance with the applicable drawing.

3.6.1 Electrical resistance. -When tested electrically, as prescribed in Para. 4.4.1.1, the electrical resistance of the ignition element assembly shall be in accordance with the applicable drawing.

3.7 Electrical sensitivity. -When tested as prescribed in Para. 4.4.2, the ignition element assembly shall not function when subjected to electrical current of 2.00 $\pm$ .05 ampere. Ignition element assembly shall function when subjected to an electrical current of 4.00 $\pm$ .20 amperes.

3.8 Electrical functioning. -The ignition element assembly shall function within a period of time no greater than 6.00 milliseconds when tested as prescribed in the electrical functioning test of Para. 4.4.3.

3.9 Vibration. -The ignition element assembly shall be both safe and operable after having been subjected to the vibration test prescribed in MIL-STD-353, Para. 4.4.4 and 4.4.4.1.

3.10 Forty foot drop. -The ignition element assembly shall withstand the forty (40) foot drop test prescribed in MIL-STD-352, Para. 4.4.4 and 4.4.4.2.

3.11 Temperature and humidity. -The ignition element assembly shall withstand the temperature and humidity test prescribed in MIL-STD-354 and Para. 4.4.5.1. Tests for this requirement will be conducted on each inspection lot, unless otherwise specified by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot.

4.2.1 Inspection lot. -As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be determined by the Government inspector and may differe from the quantity designated in the contract cr order as a lot for production, shipment or other purpose. For the purpose of sample selection, an inspection lot size of 1,000 minimum shall be used. The Sampling Plan of Para. 4.4.2.1 gives the number of samples to be used for four representative lot sizes.

4.2.2 Preproduction lot. -Unless otherwise specified by the procuring activity, the preproduction lot shall consist of completely assembled and loaded ignition elements, of one mark and mod. The Government will inspect these elements for compliance with all of the requirements and tests prescribed in this document. If the sample proves to be acceptable, the contractor will be authorized by the procuring activity to proceed with the submission of the product for acceptance.

4.2.3 Production lot. -For procurement purposes, each lot of loaded ignition elements shall consist of 4,000 plus those required for test purposes. Unless otherwise stated in the contract, order or requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.2.1.

4.3 Inspection.

4.3.1 Place of inspection and tests. -The inspection and tests specified shall be at the plant of manufacture unless specifically designated at another location by the procuring activity.

4.3.2 Method of sample selection. -All samples shall be selected at random, using a table of random numbers. A separate listing shall be made for each Inspection Lot.

4.4 Acceptance tests. -Except where herein specified, acceptance tests shall be conducted on the basis of each lot as provided by 4.2.1 and 4.2.2. In the absence of notable variations in the fabrication procedures and product quality as established by the preproduction sample, the requirements of the drawing and conformance

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with the acceptance tests prescribed in this document will govern the acceptance of production material. Advisory provisions pertaining to acceptance testing under special conditions of procurement are contained in Para. 6.4.

4.4.1 Insulation resistance. -The resistance between Inner Cup and Electrode, prior to bridging, shall be not less than fifty (50) megohms when an alternating potential of 500 volts, 60 cycle, is applied for one second. Ignition elements shall be checked 100 percent for insulation resistance.

4.4.1.1 Electrical resistance. -Prior to checking resistance, the exterior surface of all ignition elements shall be inspected for freedom from sealant or other foreign matter. A coloring agent shall be added to render excess sealant detectable. Each ignition element assembly shall be tested for conformance with the bridge circuit resistance of the drawing by means of a test circuit which limits the bridge current to 50 milliamperes maximum. Suitable safety precautions shall be taken to protect the operator during the application of the current. If the bridge circuit resistance falls within the limits of the drawing, the assembly shall be acceptable. Ignition elements shall be checked 100 percent for electrical resistance.

4.4.2 Electrical sensitivity test. -Electrical sensitivity shall be ascertained by acceptable performance in the non-firing test of Para. 4.4.2.2 and the firing test of Para. 4.4.2.3.

4.4.2.1 Sampling plan. -Tests outlined shall be in accordance with MIL-STD-105 and MIL-STD-414. Ignition elements of Para. 4.4.2.2 and those of Para. 4.4.2.3 may be used to perform test Para. 4.4.2.4.

SAMPLING PLAN

This table gives the sample quantities for the various tests for commonly-used inspection lot sizes. The sample quantities for other lot sizes may be taken from MIL-STD-105 and MIL-STD-414.

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SAMPLING PLAN

PARAGRAPH	TEST	INSPECTION LOT 0-1300	1301-3200	3201-8000	8001-22000
4.4.2.2	Non-Firing	110	110	110	110
4.4.2.3	Firing	110	110	110	110
4.4.2.4	Output	110	110	110	110
4.4.3	Electrical Functioning	35	40	50	75
4.4.4.1	Vibration	15	25	35	50
4.4.4.2	Drop	15	25	35	50
4.4.5.1	Temperature & Humidity	25	25	25	25

4.4.2.2 Non-Firing Test. -Sample ignition elements shall be selected from each inspection lot in accordance with MIL-STD-105, Inspection level L-8, AQL, 0.10%. Each element shall be mounted in the firing test fixture, BuWeps L.D. 546657. Apply an open circuit D.C. potential of 24 volts to the test firing fixture terminals. Include in the firing circuit, series resistances sufficient to regulate the current to 2.00 $\pm$ .05 amperes. These resistances should be variable to adjust for the varying resistances of the ignition elements. An ignition element which fires is to be considered a defective. One or more defectives will cause rejection of the lot.

4.4.2.3 Firing Test. -Sample ignition elements shall be selected from each inspection lot in accordance with MIL-STD-105, Inspection level L-8, AQL 0.10%. Each element shall be mounted in Firing Test Fixture, BUWEPS LD 546657. Apply an open circuit DC potential of 24 volts to the test firing fixture terminals in accordance with Dwg. 2302106. A series resistor sufficient to regulate the current to 4.00 $\pm$ .20 amperes shall form a part of the electrical circuit. An ignition element which fails to fire shall be considered defective. Since poor contacts within the firing circuit can cause apparent failure, it is recommended that prior to applying the firing current the resistance of each item be checked. Re-tests: The items in the firing test are concurrently tested for output as part of the output test. If, at the completion of the firing test, the number of defectives equal or exceed the rejection number, the remainder of the output test shall be concurrently given this firing test.

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4.4.2.4 Firing test output. -Sample ignition elements shall be selected from each production lot in accordance with MIL-STD-105. Inspection Level L-8, AQL 0.10% defective. Each element shall be mounted in Firing Test Fixture, BUWEPS LD 546657, and an open circuit DC potential of 24 volts applied to the test fixture terminals. The ignition element to be tested must be capable of puncturing a brass diaphragm of .008 thickness, Dwg. 2302112, to successfully ignite a quantity of Black Powder JAN-P-223-a Grade FFG. The diaphragm shall be located 2.0 from the face of the ignition element closure disc when assembled in the test fixture. The test charge of 0.140 grams approximately of Black Powder shall be in intimate contact with the diaphragm. Ignition elements failing to puncture diaphragm or ignite test charge shall be considered defective.

4.4.3 Electrical functioning test. -The ignition element assembly shall be mounted in Firing Test Fixture, BUWEPS LD 546657. Apply an open circuit DC potential of 24 volts to the firing test fixture terminals in accordance with Dwg. 2302106. It is important that firm contact on clean bright surfaces is obtained to avoid high contact resistance. The DC voltage shall be supplied by an essentially non-inductive source of 2.0 ohms internal resistance or by means of a circuit of equivalent characteristics. The firing time shall be measured using an electronic timer which is actuated by the firing pulse and the explosive output. The response from the explosion shall be detected from the blast at a distance less than 1.0 from the closing disc of the ignition element. An ionization probe, or a sensing device of at least equal response time, shall be used for blast detection. Acceptability shall be based on functioning time within a period no greater than 6.0 milliseconds. Sampling and disposition shall be on the basis of each production lot in accordance with MIL-STD-414, Inspection Level IV, AQL 0.10% defective, using an upper specification limit of 6.0 milliseconds. If one element firing time greater than 6.0 milliseconds occurs, it shall be disregarded and another element shall be tested in its place. If one element fails to fire, or if two or more firing times greater than 6.0 milliseconds occur, the lot shall be rejected. NOTE: Allowable instrumentation error to be maintained at 2 percent maximum.

4.4.4 Durability tests. -Durability shall be ascertained by acceptable performance in the vibration test of Para. 4.4.4.1 and the forty foot drop test of Para. 4.4.4.2. These tests shall

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be conducted on the first five consecutive production lots, and on any one of each five consecutive lots thereafter. A lot immediately subsequent to a reject lot shall be considered a first lot.

4.4.4.1 Vibration Test. -From each lot select a sample in accordance with MIL-STD-105, Inspection level L-6, acceptance number zero. Sample elements from each production lot submitted for test shall be mounted in Vibration Test Fixture, BUWEPS LD 546651, as shown on the applicableassembly drawing. The fixture shall be rigidly mounted to the vibration table prescribed in MIL-STD-353. The ignition elements shall be vibrated in each of the following positions: (1) axis vertical, closing disc up; (2) axis vertical, closing disc down; and (3) axis horizontal. The vibration test shall be conducted at a single amplitude of 0.030 inch throughout a frequency range of 1100 to 3000 cycles per minute in frequency steps of 100 cycles per minute with a running time of 10 minutes at each increment. Each element shall receive a total of 10 hours of vibration in the test when vibrated for 200 minutes at each position. Except as stated above, equipment and test procedure shall be as prescribed in MIL-STD-353. Having been subjected to the vibration test, the ignition element shall be both operable and safe to handle or use. To establish operability, the tested ignition element shall be subjected to the Electrical Functioning Test of 4.4.3. The firing time of each element shall be recorded. The criteria for acceptance is that this data not be significantly different from data obtained in Electrical Functioning Test. An ignition element which fails to pass the Vibration Test as herein prescribed shall be considered defective. One or more defectives shall be cause for rejection of the lot.

4.4.4.2 Forty Foot Drop Test. -From each lot select a sample in accordance with MIL-STD-105, Inspection level L-6, acceptance number zero. Sample ignition element assemblies from each production lot submitted for test shall be subjected to the Forty Foot Drop Test. Sample elements shall be assembled in Drop Test Fixture, BUWEPS LD 546658, shown on the assembly drawings for the fixture, preparatory to being drop tested. Five of the ignition elements shall be dropped to strike axis vertical, closing disc down; five, axis vertical, closing disc up; and five, axis horizontal. The test procedure and equipment used in conducting the test shall conform with that prescribed in MIL-STD-352. The criteria by which the ignition element is judged to have withstood this test are that (1) the element shall be both safe to handle and dispose of, and (2) no explosive element shall have become initiated. An ignition element which fails to pass the Forty Foot Drop Test as herein prescribed shall be considered defective. One or more defectives shall be cause for rejection of the lot.

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4.4.5 Preproduction Sample Tests. -In addition to the acceptance tests of Para. 4.4.1 through 4.4.4.2, the preproduction sample shall be subjected to the Temperature and Humidity Test of Para. 4.4.5.1. Conformance of the preproduction sample, where an inspection level is provided, shall be ascertained on sample sizes as provided for a lot size of 4,000.

4.4.5.1 Temperature and Humidity Test. -Twenty-five ignition element assemblies shall be selected from each inspection lot. Sample elements shall be subjected to the Temperature and Humidity Test prescribed in MIL-STD-354. Elements shall be positioned vertically to the test chamber and supported in a manner to prevent submersion. The output end of element shall face downward throughout the test. Having been subjected to the test, the elements shall be both operable and safe to handle or use. To establish operability, the tested ignition elements shall be subjected to the Electrical Functioning Test of Para. 4.4.3. An ignition element which fails to pass this test shall be considered a defective. One or more defectives shall cause rejection of the lot. Following the cycling, each element shall be checked for continuity. They shall then be fired and the firing times recorded, using the equipment and methods of the Electric Functioning Test, as prescribed in Para. 4.4.3. The criteria for acceptance is that this data shall not be significantly different from that obtained in the Electric Functioning Test. Elements which fail to pass this test shall be considered defective. One or more defectives shall cause rejection of the lot. The test of MIL-STD-354 requires 28 days of cycling in a set sequence. However, for testing, these elements may be inserted into the test chamber and removed at any time during the cycling as long as all the cycles are performed.

4.5 Test Results. -Test results of the Non-Firing Test (Para. 4.4.2.2), the Firing Test (Para. 4.4.2.3), the Electrical Functioning Test (Para. 4.4.3) and the Durability Tests (Para. 4.4.4) shall be forwarded to the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.

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5.1.1.1 One thousand (1000) ignition elements of one mark and mod shall be packaged in accordance with BUWEPS LD No. 546664.

5.1.2 Level C.

5.1.2.1 One thousand (1000) ignition elements of one mark and mod shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Pack four (4) inner containers, BUWEPS LD 546664 in accordance with BUWEPS LD 482723.

5.2.2 Level C.

5.2.2.1 Four (4) inner containers shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to Code of Federal Regulations 49 CFR 71-78.

5.3 Marking.

5.3.1 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data Cards. -Data cards, five inches by eight inches, printed as shown below, shall be completed for each lot of elements. Two of these shall accompany the shipping ticket. Three copies shall be forwarded to the Bureau of Naval Weapons; one copy shall be forwarded to OSO, Mechanicsburg, Pennsylvania (Code AM); one copy shall be forwarded to NAD, Crane, Indiana (Code 3500); and one copy shall be forwarded to the test activity to represent each group of samples submitted for test.

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IGNITION ELEMENT, ELECTRIC, MK 10 MOD 0

Preproduction/Production Lot Number_____(Delete one)_____

Quantity in Lot_____ Name of Manufacturer_____

List of Drawings_____ Contract Number_____

Specification Number_____

Lot Produced_____(Month and Year)_____

Remarks:

(Enter a description of each waiver, deviation, or both, related to the lot represented with reference to the authorization for the waiver or deviation.)

6. NOTES

6.1 Intended use. -The Ignition Element Mk 10 Mod 0 is intended for use in the ignition of the main propellant charge in the Mk 33-1 Impulse Cartridge.

6.2 Ordering data. - Procurement documents should specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

6.3 Advisory procedures for preparation and loading of ignition charge.

6.3.1 Preparation of ignition charge. -A small amount, approximately 10 milliliters of the watered lead styphnate slurry, as received is placed in a tared rubber dish. The styphnate is allowed

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to settle, and the supernatant water is removed by decantation or by siphoning. Approximately 25 milliliters of ethanol are added, and the mixture is stirred gently, allowed to settle, and siphoned. An additional 25 milliliters of ethanol are added, and the stirring, settling and siphoning procedure repeated. The remaining mixture is airdried at ambient temperature for two (2) hours and then oven-dried for twelve (12) hours at 90° F. The dry weight of the styphnate is then determined. In a separate container, dry cellulose nitrate, equivalent to 2.0 percent of the dry weight of the lead styphnate, is mixed with butyl acetate to make an approximately 1/2 percent solution. (A weighted quantity of gravimetrically standardized stock solution of approximately 1/2 percent cellulose nitrate concentration may be used to obtain the 2.0 percent of cellulose nitrate required.) This solution is added to the dry styphnate and mixed until the styphnate is thoroughly wetted.

6.3.2 Segregation and handling of ignition charge. -The ignition charge is transferred to loading dishes in two (2) gram batches. The loading dish should be constantly checked to prevent the mixture from becoming too dry, particularly around the edges and on top of the dish. If necessary, the operator may add butyl acetate and work the mixture to the proper buttering consistency. A maximum of two (2) grams of mix should be present at each loading station, and then only when an operator is present. Any of the charge which is spilled should be cleaned up immediately by means of a cloth which has been wet with butyl acetate. The soiled cloth should be placed in a special explosives waste disposal can.

6.3.3 Operating precautions.

(a) Dry lead styphnate is characterized by unusual sensitivity to electrical discharge. Except when absolutely necessary to obtain accurate weights of lead styphnate, it should never be handled in a dry state. Bulk lead styphnate should be stored under water or butyl acetate. All loading, mixing and handling of lead styphnate or lead styphnate ignition charge should be done in rooms kept at a relative humidity of at least 80 percent.

(b) All mixing and loading of the ignition charge should be done behind approved safety shield whenever practical. All fixtures, equipment, benches and floors should be properly grounded to prevent accidental ignition due to static discharge. This applies to spatulas used in buttering operations which should be grounded by means of a grounding wire connected directly to a grounded bus bar.

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(c) Butyl acetate confers a degree of resistance to explosion and ignition by both impact and friction. However, it is emphasized that the resistance to explosion obtained by wetting with butyl acetate is of a lower order than that obtained by wetting with water. Butyl acetate is flammable and produces vapors which in themselves can be an explosive and fire hazard.

6.4 Advisory acceptance provisions under special procurement conditions. -When the Ignition Element, Electric, Mk 10 Mod 0, is not an item of prime procurement but is manufactured as a component part of, and for inclusion in, a specific application which is an item of prime procurement, acceptance tests for conformance with the requirements of 3.7, 3.8, 3.9, 3.10 and 3.11 may be waived by the Government and acceptance made on the basis of similar prescribed tests on the complete subject. Tests in accordance with the prime specification may be used only to justify acceptance of the ignition element, but not rejection. Rejection of the ignition element will be made only on the basis of the acceptance tests prescribed in this document. The provisions provided herein are intended for convenience in Government inspection, and do not absolve the prime contractor of his responsibility to manufacture ignition elements which comply with all of the requirements of this document.

6.5 General safety precautions. -The loading, assembling and handling of the item covered by this specification, and the sub-assemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a contractor or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At the Bureau of Naval Weapons, Navy Department, and other government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail technical requirements of the specification.

6.5.1 All loading operations should be conducted in a neat and orderly manner.

6.5.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations such as, mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar

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as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.5 All explosives and completely or partially loaded items should be stored in suitable storage mazagine located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.5.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

6.5.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiating by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items

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should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulation aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (See 6.5.4) of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6.5.6.5 Protect all explosives operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completion of an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators, or squibs.

6.5.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to

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minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately. Employ an adequate ventilation system to prevent an accumulation of toxic or flammable vapors in manufacturing or storage areas.

Notice. -When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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PARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander
Naval Ordnance Systems Command Headquarters
Attn: Code 9343
Washington, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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